

Analytics for Stake-Holders

Presented by GROUP-6



Manufacturing Project

A manufacturing project is any organized effort in the manufacturing industry that aims to create, improve, or optimize products, processes, or production systems.

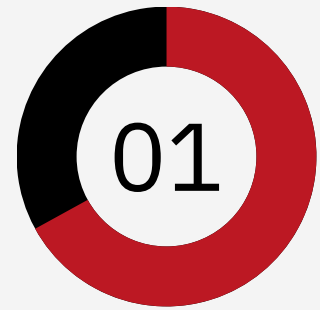


Objectives

- To analyze and monitor the overall manufacturing performance.
- Identify bottlenecks in production and rejection processes.
- Provide department-wise, employee-wise, and machine-wise insights.
- Support decision-making with real-time data visualization.

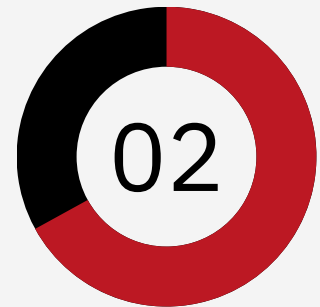


Key Insights at a Glance



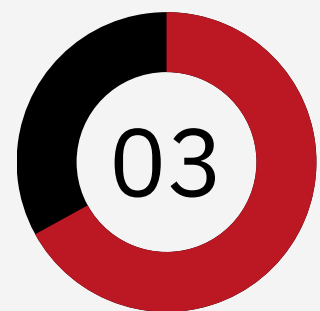
Strong Production Base

49.8M units manufactured with stable monthly output.

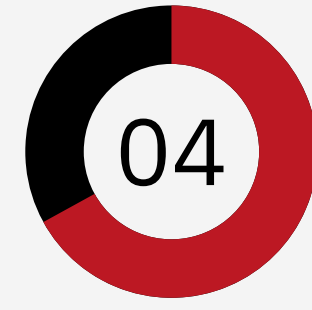


Quality Concerns

~491K rejections & ~490K wastage → ~1M units lost.

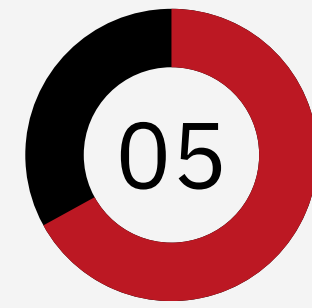


Certain departments like Knitwear & Printed Fabric show higher rejection rates.



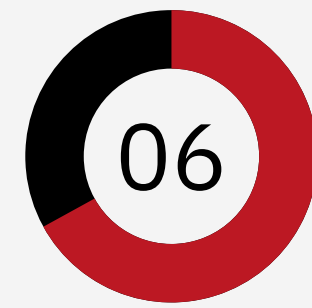
High Dependence on Key Buyers

Uniqlo, Zara, H&M & Nike dominate purchases.



Operational Bottlenecks

Few employees & machines account for major rejections.



Seasonal Variations

April peaks in production, February remains lowest.

Overall Production Snapshot

Total
Manufactured
Quantity

49.79M

Total
Processed
Quantity

60.02M

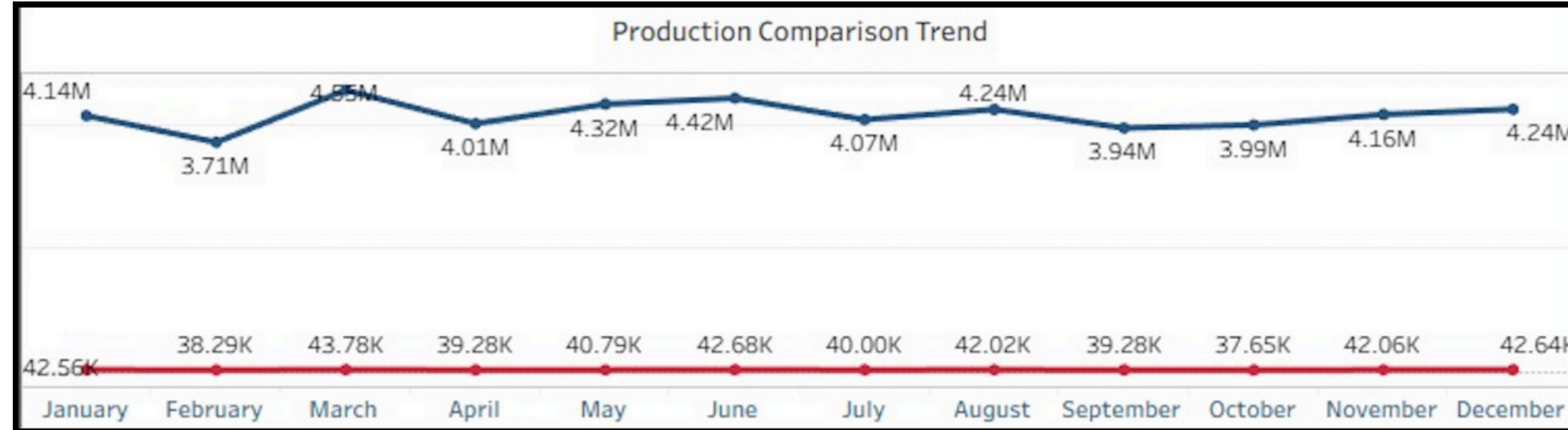
Total
Rejected
Quantity

491.02K

Total
Wastage
Quantity

490.29K

Production Trend Insights



- **Steady Output:** Production ranges 3.7M–4.7M units per month.
- **April Peak:** Highest production at 4.01M units.
- **February Dip:** Lowest at 3.71M units.
- **Rejection Stability:** Despite production changes, rejections remain steady at ~40K/month.

Machine Efficiency

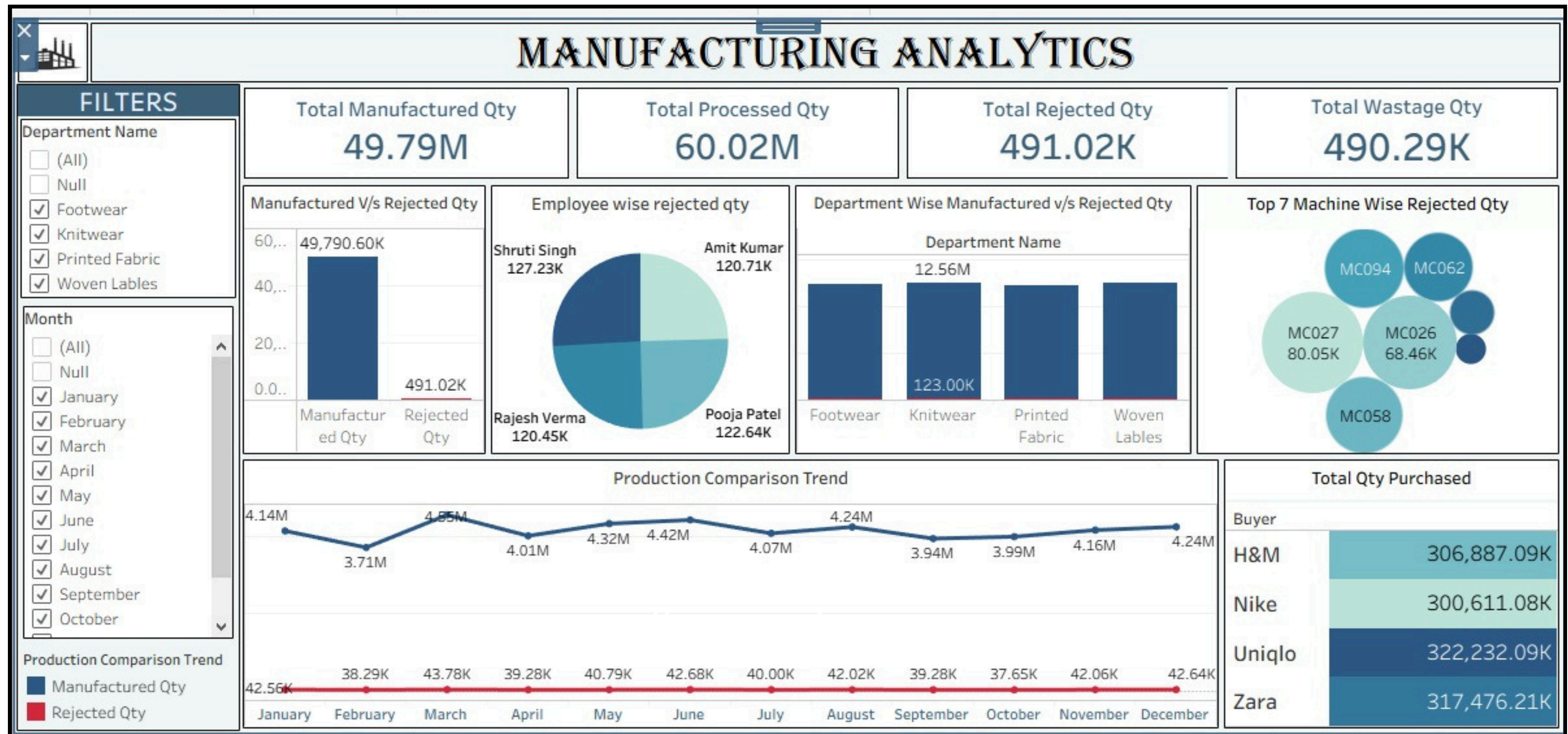
- **Key Outliers:** MC094, MC027, MC058, MC026 & MC062 are major rejection sources.
- **MC027 Alone:** Responsible for 80K rejected units.
- **Overall Impact:** Few faulty machines causing disproportionate losses.



Prioritize preventive maintenance & replacement → direct impact on reducing scrap.

Top-7 Machine-wise Rejected Qty





Total Manufactured Qty

49.79M

Total Processed Qty

60.02M

Total Rejected Qty

491.02K

Total Wastage Qty

490.29K

Manufactured V/s Rejected Qty

49,790.60K

491.02K

Manufactur
ed Qty

Rejected
Qty

Employee wise rejected qty

Shruti Singh
127.23K

Amit Kumar
120.71K

Rajesh Verma
120.45K

Pooja Patel
122.64K

Department Wise Manufactured v/s Rejected Qty

Department Name

12.56M

123.00K

Footwear

Knitwear

Printed
Fabric

Woven
Lables

Top 7 Machine Wise Rejected Qty

MC094

MC062

MC027
80.05K

MC026
68.46K

MC058

Production Comparison Trend

4.14M

3.71M

4.85M

4.01M

4.32M

4.42M

4.07M

4.24M

3.94M

3.99M

4.16M

4.24M

42.56K

38.29K

43.78K

39.28K

40.79K

42.68K

40.00K

42.02K

39.28K

37.65K

42.06K

42.64K

January

February

March

April

May

June

July

August

September

October

November

December

Total Qty Purchased

Buyer

H&M

Nike

Uniqlo

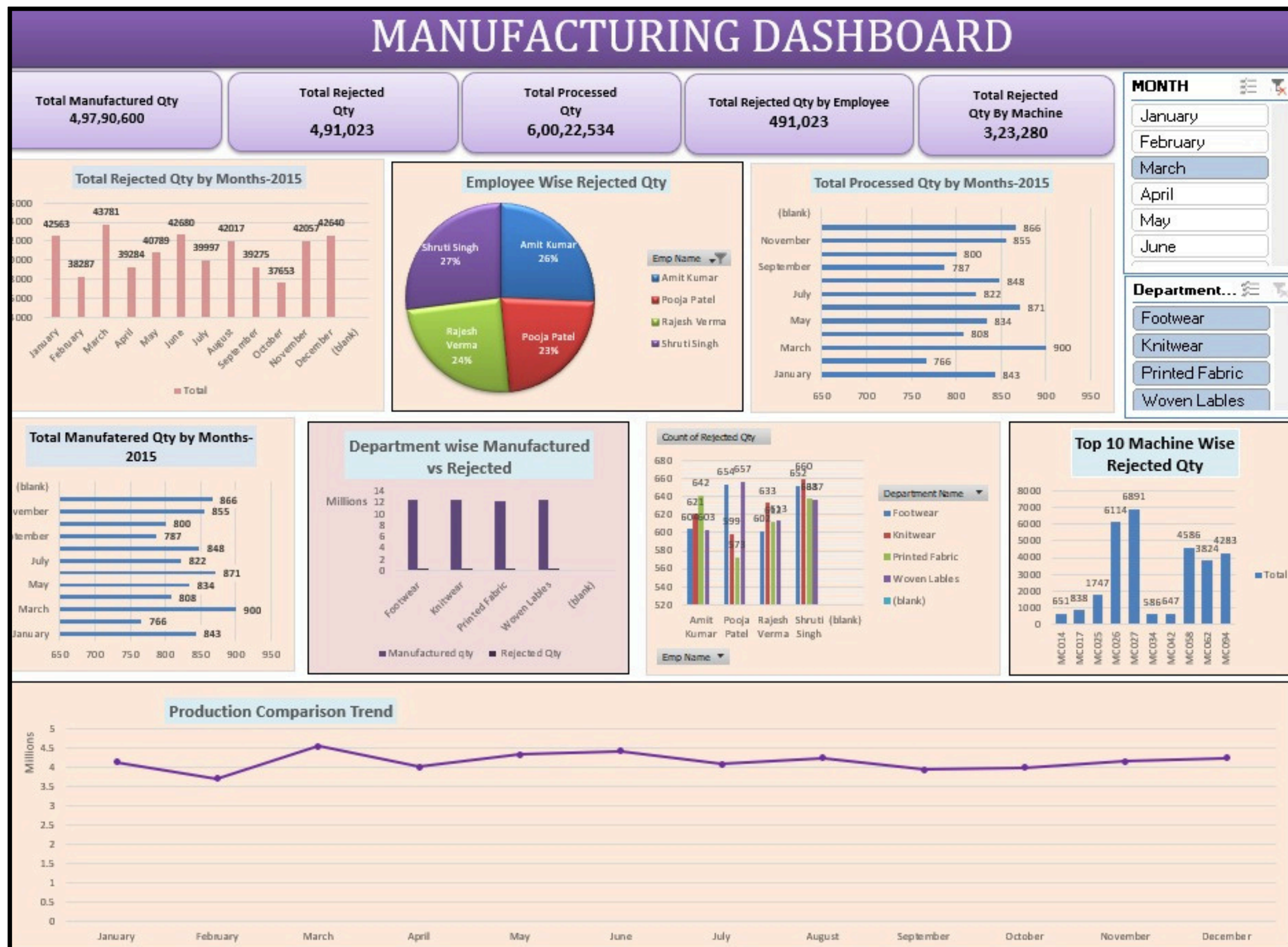
Zara

306,887.09K

300,611.08K

322,232.09K

317,476.21K





MANUFACTURING DASHBOARD

50M

Total Manufactured Qty

491K

Total Rejected Qty

60M

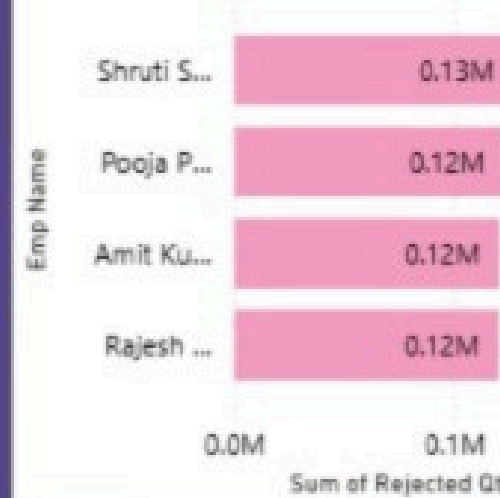
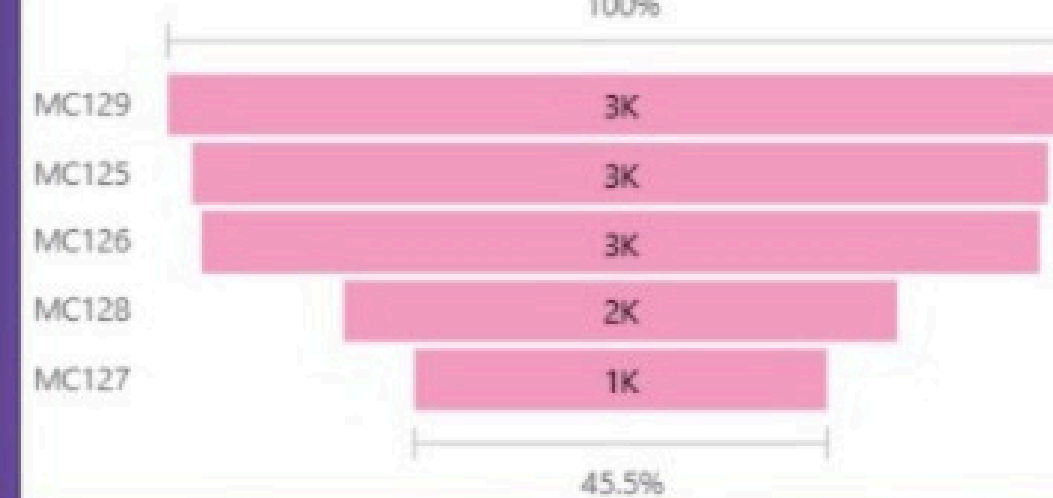
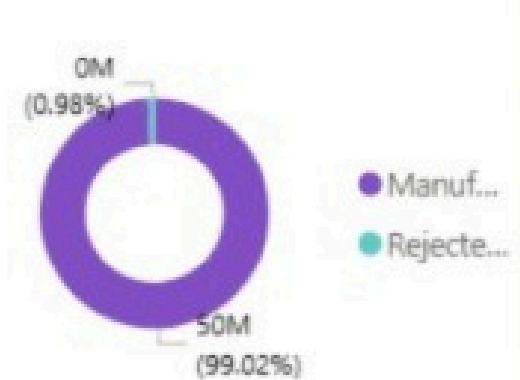
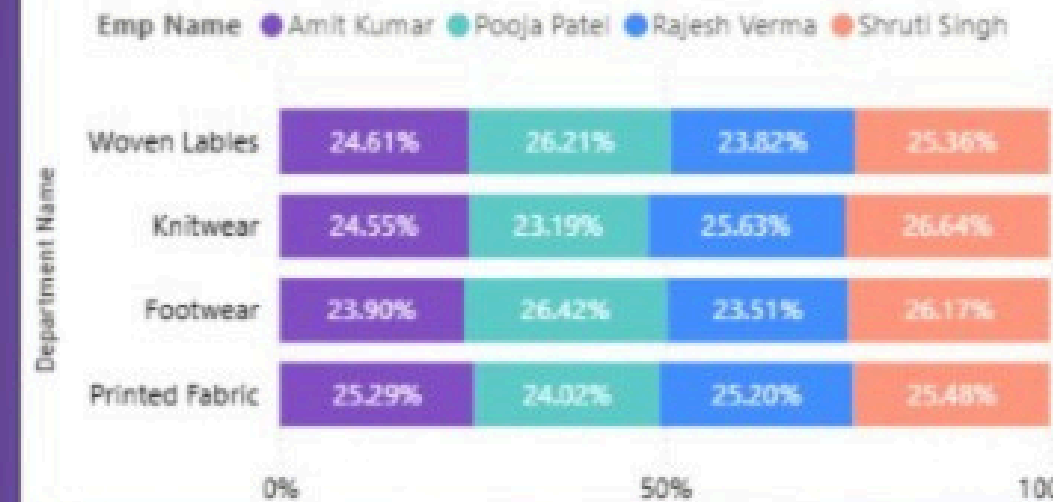
Total Processed Qty

1bn

Total Qty

75M

Total WO Qty

Employee wise Manufactured Qty**Employee wise Rejected Qty****Top 5 Machine wise Rejected Qty****Department wise Manufactured V/S Rejected Qty****Manufactured V/S Rejected Qty****Department wise Employee wise Rejected Qty****Machine Code**

- ☐ MC001
- ☐ MC0010
- ☐ MC002
- ☐ MC003
- ☐ MC004
- ☐ MC005
- ☐ MC006
- ☐ MC007

Year, Month

- ☐ 2015
- ☐ January
- ☐ February
- ☐ March

Department

- ☐ Footwear
- ☐ Knitwear
- ☐ Printed Fabric
- ☐ Woven Lables

Emp Name

- ☐ Amit Kumar
- ☐ Pooja Patel
- ☐ Rajesh Verma
- ☐ Shruti Singh

A GLIMPSE OF SQL

```
31  #Employee wise rejected qty
32
33  •  SELECT
34      `Emp Name`,
35      ROUND(SUM(`Rejected Qty`) / 1000000, 2) AS Rejected_Qty_Millions
36  FROM manufacturing_data
37  GROUP BY `Emp Name`
38  ORDER BY Rejected_Qty_Millions DESC;
39
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	Emp Name	Rejected_Qty_Millions
▶	Shruti Singh	0.13
	Rajesh Verma	0.12
	Amit Kumar	0.12
	Pooja Patel	0.12

```
79  #Department wise manufactured v/s rejected qty
80
81  •  SELECT
82      `Department Name`,
83      ROUND(SUM(`today Manufactured qty`) / 1000, 2) AS Manufactured_Thousands,
84      ROUND(SUM(`Rejected Qty`) / 1000, 2) AS Rejected_Thousands
85  FROM manufacturing_data
86  GROUP BY `Department Name`
87  ORDER BY Manufactured_Thousands DESC;
88
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	Department Name	Manufactured_Thousands	Rejected_Thousands
▶	Woven Lables	12573.11	125.80
	Knitwear	12561.43	123.00
	Footwear	12387.53	122.86
	Printed Fabric	12268.53	119.38

**Thank you
very much!**

